PRODUCTION LINE STABILITY PREDICTION

Machine Learning-Based Case Study

Project Information	Details
Project Title	Production Line Stability Prediction
Application Area	Manufacturing / Industrial Machine Learning
Primary Tool	Altair AI Studio (RapidMiner) 2026.1.1
Model	Random Forest Classification
Validation	5-Fold Stratified Cross-Validation
Optimization	Grid-Based Hyperparameter Optimization
Dataset	AI4I 2020 Predictive Maintenance Dataset
Project Context	SynTech 2026 – Data Science Contest

1. Abstract

This case study presents a machine learning-based approach for predicting production line stability from manufacturing process data. The project was developed using Altair AI Studio (RapidMiner) and the AI4I 2020 Predictive Maintenance Dataset. The workflow includes data preparation, attribute selection, creation of a production stability target, model training, cross-validation, and grid-based hyperparameter optimization. A Random Forest classifier was used to classify production conditions as Stable or Unstable. The optimized workflow achieved an overall cross-validated accuracy of 98.68%. The results also show why class-specific evaluation is important because unstable observations are much less frequent than stable observations.

2. Introduction

Manufacturing systems continuously generate process measurements such as temperature, rotational speed, torque, and tool wear. Analysing these measurements can help identify operating conditions associated with machine failures and production instability. Machine learning provides a way to learn patterns from historical process data and classify new operating conditions.

The project was developed as part of the SynTech 2026 Data Science Contest and is being retained as the basis for the final project. The current implementation covers the machine learning workflow and its evaluation. A public API or Mendix integration is not claimed as part of the completed implementation.

3. Problem Statement

Production line instability can affect manufacturing efficiency and may be difficult to identify early from process data. This project aims to use machine learning to predict stable and unstable production conditions from manufacturing process measurements.

4. Objectives

- Prepare and analyse manufacturing process data for machine learning.
- Create a production stability classification target from the available machine-failure information.
- Select relevant attributes and remove identifiers and unsuitable variables.
- Develop a Random Forest classification model using Altair AI Studio.
- Apply 5-fold stratified cross-validation.
- Use grid-based hyperparameter optimization to examine different Random Forest configurations.
- Evaluate accuracy, precision, and recall for Stable and Unstable classes.
- Identify limitations and practical directions for extending the project.

5. Dataset Description

The project uses the AI4I 2020 Predictive Maintenance Dataset. It contains 10,000 manufacturing observations and 14 original attributes covering product information, process measurements, machine failure, and failure-mode indicators. The dataset is synthetic and is intended for predictive-maintenance research and educational use.

Characteristic	Value
Observations	10,000

Original attributes	14
Product type	L: 6,000; M: 2,997; H: 1,003
Machine failure = 0	9,661
Machine failure = 1	339
Failure proportion	3.39%
Missing values	None reported in the project dataset

Main modelling attributes include Air temperature [K], Process temperature [K], Rotational speed [rpm], Torque [Nm], Tool wear [min], Type, and the derived Process Stability target.

6. Target Variable and Data Preparation

The dataset does not contain an independent operational label named Stable or Unstable. For this project, Process Stability was derived from Machine failure: Machine failure = 0 was assigned Stable and Machine failure = 1 was assigned Unstable.

Identifier-type attributes and failure-mode indicator attributes were removed during preparation. After the stability target was generated, the original Machine failure attribute was removed so that it was not retained as a direct input feature.

Stage	Action
Data input	Read the AI4I 2020 CSV dataset
Attribute selection	Remove identifiers and unsuitable attributes
Target creation	Generate Process Stability from Machine failure
Role assignment	Set Process Stability as the label
Final selection	Retain modelling attributes and remove the original Machine failure field

7. Machine Learning Workflow

The completed Altair AI Studio process follows a sequence from data loading through model optimization and evaluation.

1. Read CSV — load the manufacturing dataset.
2. Select Attributes — remove attributes not used for modelling.
3. Generate Attributes — create Process Stability.
4. Set Role — assign Process Stability as the label.
5. Select Attributes (2) — perform final feature selection.
6. Optimize Parameters (Grid) — evaluate Random Forest hyperparameter combinations.
7. Cross Validation — perform stratified 5-fold validation.
8. Random Forest — train the classifier within each training fold.
9. Apply Model — apply the trained model to the validation fold.
10. Performance — calculate binominal classification performance.

Workflow: Read CSV → Select Attributes → Generate Attributes → Set Role → Select Attributes → Optimize Parameters (Grid) → Cross Validation → Random Forest → Apply Model → Performance

8. Model Development

Random Forest was selected as the main classification algorithm. It combines multiple decision trees to produce a classification result. In this project, the model was trained inside Cross Validation so that evaluation is performed on validation folds rather than on the same observations used for training.

The recorded tree visualisation uses process variables such as Torque, Tool wear, Rotational speed, and Air temperature in decision rules, providing an interpretable view of how individual trees separate Stable and Unstable observations.

9. Hyperparameter Optimization

Grid-based optimization was used to examine different Random Forest configurations. The selected parameters were number of trees, maximum depth, and classification criterion.

Parameter	Values Evaluated

Number of trees	50, 100, 150, 200
Maximum depth	5, 10, 15, 20
Criterion	Information gain, Gini index, and other configured classification criteria
Total combinations	48

The optimization results contained 48 parameter combinations. The recorded optimization table reached approximately 0.987 accuracy for its strongest configurations, while the final evaluated optimized workflow produced 98.68% overall accuracy in 5-fold stratified cross-validation.

10. Validation Strategy

The model was evaluated using 5-fold stratified cross-validation. Stratification is important because the Stable and Unstable classes are highly imbalanced. Each fold therefore attempts to preserve the class distribution during training and validation.

Setting	Configuration
Validation method	Cross Validation
Number of folds	5
Sampling	Stratified sampling
Model	Random Forest
Evaluation	Binominal classification performance

11. Results

The final recorded performance result from the optimized workflow is shown below.

Metric	Stable	Unstable
Precision	98.90%	90.27%
Recall	99.74%	68.44%

Overall Metric	Result
Accuracy	98.68% ± 0.10%
Micro-average accuracy	98.68%

Recorded confusion matrix:

Prediction / Actual	True Stable	True Unstable
Predicted Stable	9,636	107
Predicted Unstable	25	232

12. Result Interpretation

The overall accuracy of 98.68% indicates that the classifier correctly classified most observations in the validation process. Stable-class recall is 99.74%, meaning that most Stable observations were correctly identified.

Unstable-class recall is 68.44%. Therefore, a portion of unstable observations were classified as Stable. For a production-monitoring application, this metric deserves particular attention because an undetected unstable condition may be important to investigate.

The dataset is strongly imbalanced, so accuracy alone should not be treated as the only measure of model quality. Future versions should explicitly examine Unstable recall, precision, F1-score, and precision-recall behaviour.

13. Key Outcomes

- Built a complete machine learning workflow in Altair AI Studio.
- Used Random Forest classification for production stability prediction.
- Applied stratified 5-fold cross-validation.
- Evaluated 48 configured hyperparameter combinations.
- Achieved 98.68% overall cross-validated accuracy in the recorded final result.
- Obtained 90.27% precision for the Unstable class.

- Identified Unstable-class recall as the principal area for further improvement.

14. Limitations

- The AI4I 2020 dataset is synthetic rather than a direct record from a specific industrial production line.
- The Stable/Unstable target is derived from Machine failure and is therefore not an independent operational stability label.
- The dataset is strongly imbalanced, with Unstable observations representing 3.39% of the data.
- The current implementation is a local Altair AI Studio workflow and does not claim a deployed public API or Mendix integration.
- Further imbalance-aware evaluation can strengthen the project.

15. Future Scope

- Improve minority-class detection using class weighting or resampling methods.
- Compare Random Forest with Logistic Regression, Decision Tree, SVM, or Gradient Boosting.
- Perform threshold analysis to study the trade-off between Unstable recall and precision.
- Add feature-importance and explainability analysis.
- Develop a user-facing prediction interface for production parameters.
- Integrate the validated model with a web application such as Mendix through an appropriate deployment/API layer when the required hosting capability is available.
- Evaluate the approach on real industrial production data when available.

16. Tools and Technologies

Component	Technology
Data Science Platform	Altair AI Studio (RapidMiner)
Machine Learning	Random Forest Classification
Validation	5-Fold Stratified Cross-Validation
Optimization	Grid-Based Hyperparameter Optimization
Dataset	AI4I 2020 Predictive Maintenance Dataset
Application Layer	Mendix — future integration scope

17. Conclusion

This project demonstrates an end-to-end machine learning workflow for production stability prediction using Altair AI Studio. The workflow prepares manufacturing data, creates a stability target, trains a Random Forest classifier, evaluates multiple hyperparameter configurations, and validates the model using stratified cross-validation. The final recorded model achieved 98.68% overall accuracy, with strong Stable-class performance and comparatively lower Unstable-class recall.

The project provides a practical foundation for extending production monitoring toward an application-oriented system. The next technical focus is to improve detection of the minority Unstable class and then expose the validated model through an appropriate application or deployment architecture.

18. References

11. UCI Machine Learning Repository — AI4I 2020 Predictive Maintenance Dataset, Dataset 601, DOI: 10.24432/C5HS5C.
12. Altair AI Studio / RapidMiner Documentation — process design, modelling, validation, and optimization documentation.
13. Altair RapidMiner Documentation — deployment and web/API integration documentation.

Reference URLs:

https://archive.ics.uci.edu/dataset/601/ai4i

https://docs.rapidminer.com/